

# LEOPOLDT AND THE $p$ -LOCAL BOST-CONNES GROUPOID: THE RANK OF THE ISOTROPY, NOT ITS TRIVIALITY, AND WHY $\log_p(\mathbb{N}\mathfrak{p})$ VANISHES

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ABSTRACT. Paper XXII of the series identified two genuine points of contact between the localized Bost–Connes systems and  $p$ -adic arithmetic. We develop both and find that each of the three theorems proposed on that basis needs correction, in each case to a statement that is true and cleaner.

*Leopoldt does not make the intermediate isotropy trivial — and the isotropy is not the group the proposal computed.* By Proposition 2.2 of Paper II of the series, the isotropy at the stratum  $Y_T$  is the *divisor image*  $\operatorname{div}(\overline{K_{S,+}^\times} \cap \mathbb{I}_T)$ , and we identify it exactly:  $\operatorname{Iso}(Y_T) \cong \ker(\varphi : L_T \rightarrow U_{T^c}/\pi_{T^c}(\overline{\mathcal{O}_{K,+}^\times}))$ , where  $L_T$  is the rank- $|T|$  lattice of  $T$ -supported principal divisors with totally positive generator and  $\pi_{T^c}$  is the projection of the closed unit group to the complementary factors. The group  $\overline{\mathcal{O}_{K,+}^\times} \cap \prod_{\mathfrak{p} \in T} \mathcal{O}_{\mathfrak{p}}^\times$  that the proposal analyzed is the *kernel* of the divisor map — it contributes the identity ideal — though its rank,  $(r - \delta_p(K)) - \operatorname{rank} \pi_{T^c}$ , is exactly what budgets  $\operatorname{rank} \pi_{T^c}$ . The consequence is stronger than the proposed correction: whenever  $d_{T^c} = \sum_{\mathfrak{p} \notin T} [K_{\mathfrak{p}} : \mathbb{Q}_p] = 1$  and  $r \geq 1$ , the image of  $\pi_{T^c}$  is open — a single coordinate of an infinite-order unit has nonzero logarithm — so the quotient is finite and  $\operatorname{Iso}(Y_T)$  has *full rank*  $|T|$ . Already for a real quadratic field with  $p$  split the intermediate isotropy is nontrivial, unconditionally. The proposed equivalence “Leopoldt  $\iff$  trivial isotropy” is therefore false; Leopoldt enters only as the budget  $\operatorname{rank} \pi_{T^c} \leq r - \delta_p(K)$ .

*The proposed  $\mathcal{L}$ -invariant is identically zero.* The derivative computation is correct:  $E_p(\beta) = \prod_{\mathfrak{p}|p} (1 - \mathbb{N}\mathfrak{p}^{-\beta})$  vanishes to order  $g = \#\{\mathfrak{p} \mid p\}$  at  $\beta = 0$  with  $\frac{1}{g!} E_p^{(g)}(0) = \prod_{\mathfrak{p}|p} \log \mathbb{N}\mathfrak{p} = (\log p)^g \prod_{\mathfrak{p}} f_{\mathfrak{p}}$ . But this is the *real* logarithm. Its  $p$ -adic counterpart vanishes: Iwasawa’s branch has  $\log_p(p) = 0$ , so  $\log_p(\mathbb{N}\mathfrak{p}) = f_{\mathfrak{p}} \log_p(p) = 0$  for every  $\mathfrak{p} \mid p$  and the proposed  $\mathcal{L}_p(K) = \prod_{\mathfrak{p}} \log_p(\mathbb{N}\mathfrak{p})$  is 0. It cannot be a Fontaine–Mazur invariant.

*The regulator is a dimension, not a volume.* On a compact group Haar measure never degenerates, and  $\overline{\mathcal{O}_{K,+}^\times}$  is infinite, so the quotient has no informative volume. What is meaningful is

$$\dim(U_p/\overline{\mathcal{O}_{K,+}^\times}) = [K : \mathbb{Q}] - (r - \delta_p(K)) = r_2 + 1 + \delta_p(K),$$

so that Leopoldt is equivalent to this dimension being exactly  $r_2 + 1$  — recovering Proposition 5.3 of Paper XXII.

What survives intact is the order of vanishing:  $E_p$  vanishes at  $\beta = 0$  to order exactly  $g$ , and  $g$  is the number of exceptional zeros in the Mazur–Tate–Teitelbaum sense. That part of the analogy is correct and is the one worth keeping.

## 1. THE ISOTROPY OF THE INTERMEDIATE STRATA

Let  $K$  be a number field,  $p$  a rational prime,  $S = \{\mathfrak{p} \mid p\}$  with  $g = \#S$ , and  $U_p = \prod_{\mathfrak{p}|p} \mathcal{O}_{\mathfrak{p}}^\times$ . Write  $r = r_1 + r_2 - 1$  for the unit rank,  $\delta_p(K) \geq 0$  for the Leopoldt defect, so that  $\operatorname{rank}_{\mathbb{Z}_p} \overline{\mathcal{O}_{K,+}^\times} = r - \delta_p(K)$  [4] —  $\delta_p(K) = 0$  is known for abelian  $K$  by Baker–Brumer [1] — and for  $T \subseteq S$  put  $d_{T^c} = \sum_{\mathfrak{p} \notin T} [K_{\mathfrak{p}} : \mathbb{Q}_p]$  and  $\pi_{T^c} : \overline{\mathcal{O}_{K,+}^\times} \rightarrow \prod_{\mathfrak{p} \notin T} \mathcal{O}_{\mathfrak{p}}^\times$  for the coordinate projection. The framework

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Sources, code and data for the entire series: <https://github.com/Ruqing1963/localized-bost-connes>; this paper is in [papers/XXIII-leopoldt/](#).

is that of [5]; Papers II and XXII of the series, referred to by numeral, are unpublished companions, and every fact used from them is restated or verified in place.

**Proposition 1.1** (Kernel and projection ranks). *The group  $\overline{\mathcal{O}_{K,+}^\times} \cap \prod_{\mathfrak{p} \in T} \mathcal{O}_{\mathfrak{p}}^\times = \ker \pi_{T^c}$  satisfies*

$$(1.1) \quad \text{rank}_{\mathbb{Z}_p} \ker \pi_{T^c} = (r - \delta_p(K)) - \text{rank } \pi_{T^c} \geq \max(0, r - \delta_p(K) - d_{T^c}),$$

*with equality if and only if  $\pi_{T^c}$  has maximal rank  $\min(r - \delta_p, d_{T^c})$ . Moreover  $\text{rank } \pi_{T^c} \geq 1$  whenever  $r - \delta_p \geq 1$  and  $T^c \neq \emptyset$ : a coordinate of an infinite-order unit is an infinite-order element of  $\mathcal{O}_{\mathfrak{p}'}^\times$ , its logarithm being nonzero since  $K \hookrightarrow K_{\mathfrak{p}'}$  is injective. We emphasize that this kernel is not the isotropy of the stratum: it is the divisor-zero part, mapping to the identity ideal.*

*Proof.* The  $p$ -adic logarithm embeds  $\overline{\mathcal{O}_{K,+}^\times}$ , up to finite index, as a  $\mathbb{Z}_p$ -lattice of rank  $r - \delta_p$  in  $\bigoplus_{\mathfrak{p}|p} K_{\mathfrak{p}}$ , of total dimension  $[K : \mathbb{Q}]$ ; the kernel of the coordinate projection onto  $\bigoplus_{\mathfrak{p} \notin T} K_{\mathfrak{p}}$ , of dimension  $d_{T^c}$ , has rank given by rank-nullity, which is (1.1).  $\square$

**Theorem 1.2** (The isotropy, identified; Leopoldt does not give triviality). *Let  $T \subsetneq S$  and let  $L_T \subseteq \mathbb{Z}^T$  be the finite-index sublattice of exponent vectors  $a$  with  $\prod_{\mathfrak{p} \in T} \mathfrak{p}^{a_{\mathfrak{p}}}$  principal with totally positive generator  $\alpha_a$ . By Proposition 2.2 of Paper II,  $\text{Iso}(y) = \text{div}(\overline{K_{S,+}^\times} \cap \mathbb{I}_T)$  for  $y \in Y_T$ , and this equals, exactly,*

$$\text{Iso}(Y_T) \cong \ker \left( \varphi : L_T \longrightarrow \prod_{\mathfrak{p} \notin T} \mathcal{O}_{\mathfrak{p}}^\times / \pi_{T^c}(\overline{\mathcal{O}_{K,+}^\times}) \right), \quad \varphi(a) = \alpha_a|_{T^c},$$

*well defined since a change of generator moves  $\alpha_a|_{T^c}$  by  $\pi_{T^c}(\mathcal{O}_{K,+}^\times)$ , and with  $\pi_{T^c}(\overline{\mathcal{O}_{K,+}^\times})$  closed, the closed unit group being compact. Consequently:*

- (1) *if  $\mathcal{O}_{K,+}^\times$  is finite,  $\varphi(a)$  torsion forces  $\alpha_a$  to equal a root of unity in some completion, hence in  $K$ , so  $a = 0$ : the isotropy is trivial — Paper II's unconditional case;*
- (2) *if  $r \geq 1$  and  $d_{T^c} = 1$ , the image of  $\pi_{T^c}$  is open in the one-dimensional  $\mathcal{O}_{\mathfrak{p}'}^\times$  (Proposition 1.1, last part: a nonzero  $\mathbb{Z}_p$ -multiple of a nonzero logarithm is open), the quotient is finite, and  $\text{Iso}(Y_T)$  has full rank  $|T|$ . In particular, for  $K$  real quadratic with  $p$  split and  $T$  one of the two primes, the isotropy is infinite — unconditionally; for  $K$  totally real of degree  $n$  with  $p$  split completely and  $|T^c| = 1$  it has rank  $n - 1$ .*

*The proposed equivalence*

$$\text{Leopoldt} \iff \text{Iso}(y) = \{1\} \text{ for all } T \subsetneq S$$

*is therefore false: triviality tracks the finiteness of the unit group, not Leopoldt, and fails under Leopoldt as soon as  $r \geq 1$  (with a small complementary block).*

*Proof.* For  $a \in \ker \varphi$  pick  $\beta \in \overline{\mathcal{O}_{K,+}^\times}$  with  $\pi_{T^c}(\beta) = \alpha_a|_{T^c}$ ; then  $\alpha_a \beta^{-1} \in \overline{K_{S,+}^\times} \cap \mathbb{I}_T$  has divisor  $a$ , so  $a \in \text{Iso}$ . Conversely if  $\gamma \in \overline{K_{S,+}^\times} \cap \mathbb{I}_T$  has divisor  $a$ , approximate  $\gamma$  by totally positive  $S$ -units  $\gamma_n$ ; the divisor map is locally constant, so  $v(\gamma_n) = a$  eventually, whence  $\gamma_n = \alpha_a u_n$  with  $u_n \in \mathcal{O}_{K,+}^\times$  (times torsion), and  $\pi_{T^c}(u_n) \rightarrow \alpha_a|_{T^c}^{-1}$ ; the image  $\pi_{T^c}(\overline{\mathcal{O}_{K,+}^\times})$  is compact, hence closed, so  $\alpha_a|_{T^c}^{-1}$  lies in it and  $\varphi(a) = 0$ . Items (1) and (2) follow as indicated; for the rank in (2),  $L_T$  has rank  $|T|$  and the quotient is finite, so  $\ker \varphi$  has finite index in  $L_T$ .  $\square$

*Remark 1.3* (Numerical table). With  $p$  split completely and  $|T^c| = 1$ , so  $|T| = g - 1 = n - 1$  and  $d_{T^c} = 1$ , the corrected isotropy rank is:

| $K$                 | $r$ | $ T $ | kernel rank (1.1) | rank Iso( $Y_T$ ) |
|---------------------|-----|-------|-------------------|-------------------|
| $\mathbb{Q}$        | 0   | 0     | 0                 | 0                 |
| imaginary quadratic | 0   | 1     | 0                 | 0                 |
| real quadratic      | 1   | 1     | 0                 | 1                 |
| real cubic          | 2   | 2     | 1                 | 2                 |
| real quartic        | 3   | 3     | 2                 | 3                 |
| real quintic        | 4   | 4     | 3                 | 4                 |

The last column is Theorem 1.2(2) (full rank  $|T|$  as soon as  $r \geq 1$ ), unconditional; the proposal's values were the kernel column, which is not the isotropy. Triviality holds exactly in the first two rows — the finite-unit-group cases, which are precisely those in which Paper II (its Proposition 2.2 and Remark 2.3) is unconditional.

*Remark 1.4* (Nothing in the classification depends on this). The isotropy on the strata does not enter the main results. Paper II established topological principality of  $\mathcal{G}_S$  *directly* (its Theorem 2.5), the strata being nowhere dense regardless of their isotropy, which is what Renault's machinery requires. So Theorem 1.2 corrects a proposed characterization of Leopoldt without disturbing anything downstream.

*Remark 1.5* (The residual condition, and where Leopoldt sits). For  $d_{T^c} \geq 2$  the finiteness of the quotient in Theorem 1.2 requires  $\pi_{T^c}$  to have *open* image, i.e. rank  $d_{T^c}$  — a relative nondegeneracy of the regulator matrix restricted to the columns of  $T^c$ , automatic for  $d_{T^c} = 1$  but in general neither implied by nor implying Leopoldt, which is the single statement that the full  $r \times r$   $p$ -adic regulator is nonzero. Leopoldt enters only as the budget: all projection ranks are at most  $r - \delta_p(K)$ , so a positive defect can only *lower* rank  $\pi_{T^c}$  and thereby raise the codimension of the image.

## 2. THE EXCEPTIONAL EULER FACTOR

**Proposition 2.1** (Order of vanishing and leading term). *Let  $E_p(\beta) = \zeta_{K,p}(\beta)^{-1} = \prod_{\mathfrak{p}|p} (1 - \mathbf{Np}^{-\beta})$ . Then  $E_p$  vanishes at  $\beta = 0$  to order exactly  $g = \#\{\mathfrak{p} \mid p\}$ , and*

$$(2.1) \quad \left. \frac{1}{g!} \frac{d^g}{d\beta^g} E_p(\beta) \right|_{\beta=0} = \prod_{\mathfrak{p}|p} \log \mathbf{Np} = (\log p)^g \prod_{\mathfrak{p}|p} f_{\mathfrak{p}},$$

$f_{\mathfrak{p}}$  being the residue degrees.

*Proof.*  $1 - \mathbf{Np}^{-\beta} = 1 - e^{-\beta \log \mathbf{Np}} = \beta \log \mathbf{Np} + O(\beta^2)$ , and  $\mathbf{Np} = p^{f_{\mathfrak{p}}}$ .  $\square$

**Theorem 2.2** (The proposed  $\mathcal{L}$ -invariant vanishes). *With Iwasawa's branch of the  $p$ -adic logarithm, normalized by  $\log_p(p) = 0$ , one has*

$$\log_p(\mathbf{Np}) = \log_p(p^{f_{\mathfrak{p}}}) = f_{\mathfrak{p}} \log_p(p) = 0 \quad \text{for every } \mathfrak{p} \mid p,$$

so  $\prod_{\mathfrak{p}|p} \log_p(\mathbf{Np}) = 0$ . The quantity proposed as  $\mathcal{L}_p(K)$  is identically zero and cannot be a Fontaine–Mazur or Mazur–Tate–Teitelbaum  $\mathcal{L}$ -invariant [3, 2]: those are nonzero in the proved cases — for split multiplicative elliptic curves  $\mathcal{L}_p = \log_p(q_E)/\text{ord}_p(q_E) \neq 0$  — and conjecturally always.

*Remark 2.3* (Real versus  $p$ -adic). The confusion is between two logarithms. Formula (2.1) is a computation with the *real* logarithm, coming from the archimedean dynamics  $\sigma_t(\mu_a) = \mathbf{N}a^{it}\mu_a$ , and it is correct; it evaluates to  $(\log p)^g \prod f_{\mathfrak{p}}$ , for instance 2.5903 for  $g = 2$ ,  $f = (1, 1)$ ,  $p = 5$ , and 4.1689 for  $g = 3$ ,  $f = (1, 1, 1)$ ,  $p = 5$ . The  $p$ -adic logarithm of the same quantity is 0, because  $\mathbf{Np}$  is a power of  $p$  and  $\log_p$  is normalized to kill  $p$ . The two cannot be identified, and the vanishing

is not an accident: it is the same fact as Proposition 2.1 of Paper XXII, that  $Np$  is not a  $p$ -adic unit.

*Remark 2.4* (What is right about the analogy). The order of vanishing is correct and is worth stating on its own.  $E_p$  vanishes to order  $g$  at  $\beta = 0$ , and  $g$  is exactly the number of primes above  $p$  at which the Euler factor of  $L_p(s, \chi)$  is removed, hence the number of exceptional zeros in the Mazur–Tate–Teitelbaum sense [3] for the trivial character. That is a genuine dictionary entry, and it is the one supported by Proposition 5.1 of Paper XXII: the  $p$ -local partition function corresponds to the interpolation factor. The leading coefficient of the archimedean  $E_p$  is not the  $\mathcal{L}$ -invariant of that factor.

### 3. THE REGULATOR IS A DIMENSION, NOT A VOLUME

**Proposition 3.1** (The volume statement is not well formed).  *$G_p$  is a compact group, so its Haar measure exists, is unique up to scale and is never degenerate; normalizing  $m_{G_p}(G_p) = 1$  makes any statement about its total mass vacuous. Moreover  $\overline{\mathcal{O}_{K,+}^\times}$  is infinite whenever  $r > \delta_p(K)$ , so the index  $[U_p : \overline{\mathcal{O}_{K,+}^\times}]$  is infinite and the naive relative volume is 0 independently of  $R_p(K)$ . No equivalence with  $R_p(K) \neq 0$  can be extracted from a volume.*

**Theorem 3.2** (The correct statement).

$$\dim(U_p / \overline{\mathcal{O}_{K,+}^\times}) = [K : \mathbb{Q}] - (r - \delta_p(K)) = r_2 + 1 + \delta_p(K),$$

as  $p$ -adic analytic groups. Consequently

$$\text{Leopoldt for } K \text{ at } p \iff R_p(K) \neq 0 \iff \dim(U_p / \overline{\mathcal{O}_{K,+}^\times}) = r_2 + 1.$$

*Proof.*  $\dim U_p = \sum_{p|p} [K_p : \mathbb{Q}_p] = [K : \mathbb{Q}] = r_1 + 2r_2$  and  $\dim \overline{\mathcal{O}_{K,+}^\times} = r - \delta_p$  with  $r = r_1 + r_2 - 1$ ; subtract. The equivalence is the definition of the Leopoldt defect together with the fact that  $\delta_p = 0$  iff the  $p$ -adic regulator is nonzero.  $\square$

*Example 3.3.* Under Leopoldt:  $\dim = 1$  for  $\mathbb{Q}$ , for real quadratic, for real cubic and for real quartic fields;  $\dim = 2$  for imaginary quadratic and for complex cubic fields. In each case  $\dim = r_2 + 1$ .

**Corollary 3.4** (Agreement with Paper XXII). *Theorem 3.2 is the computation*

$$\text{rank}_{\mathbb{Z}_p} \text{Gal}(M_\infty / K) = r_2 + 1 + \delta_p(K)$$

of Paper XXII (its Proposition 5.3), transported through class field theory:  $\text{Gal}(M_\infty / K)$  is the quotient of  $U_p$  by the closure of the global units, up to a finite group. The intended statement was therefore correct in substance and misstated in form, volume for dimension.

## 4. ASSESSMENT

| Proposal                                              | Status                      | Reference |
|-------------------------------------------------------|-----------------------------|-----------|
| $\text{Iso}(Y_T) \cong \ker \varphi$ (divisor image)  | <b>exact identification</b> | Thm. 1.2  |
| full rank $ T $ for $r \geq 1$ , $d_{T^c} = 1$        | <b>unconditional</b>        | Thm. 1.2  |
| Leopoldt $\iff$ trivial isotropy                      | <b>false</b>                | Thm. 1.2  |
| topological principality affected                     | <b>no</b>                   | Rem. 1.4  |
| $\text{ord}_{\beta=0} E_p = g$                        | true                        | Prop. 2.1 |
| $\frac{1}{g!} E_p^{(g)}(0) = \prod \log \mathbf{Np}$  | true, real logarithm        | Prop. 2.1 |
| $\mathcal{L}_p(K) = \prod \log_p(\mathbf{Np})$        | <b>identically 0</b>        | Thm. 2.2  |
| $g = \#\{\text{exceptional zeros}\}$                  | <b>true; keep this</b>      | Rem. 2.4  |
| $\text{Vol}_p(U_p/\mathcal{O}^\times) \asymp  R_p _p$ | <b>not well formed</b>      | Prop. 3.1 |
| $\dim = r_2 + 1 + \delta_p$                           | <b>true; the right form</b> | Thm. 3.2  |

The two bridges identified in Paper XXII are real, and both survive; what fails is the attempt to make each carry a theorem it cannot. On the unit side, Leopoldt controls the *rank* of the closed unit group and hence the *dimension* of the symmetry group, and that is the whole of its role: it does not trivialize the isotropy of the strata, and the strata do not matter for the classification in any case. On the Euler side, the  $p$ -local partition function does correspond to the interpolation factor, and the order of vanishing does count the exceptional zeros; but the leading coefficient is an archimedean quantity, and its  $p$ -adic shadow is zero because  $\mathbf{Np}$  is a power of  $p$ .

That last point is worth isolating, because it recurs. In Proposition 2.1 of Paper XXII the  $p$ -adic dynamics failed to exist at  $\mathfrak{p} \mid p$  because  $\mathbf{Np} \notin \mathbb{Z}_p^\times$ ; here the  $\mathcal{L}$ -invariant vanishes for the same reason. Every attempt to run  $p$ -adic analysis on the norms of the primes above  $p$  meets the same wall, and it is not a technical one: those norms are precisely the elements the  $p$ -adic theory is built to discard.

Two questions. Is there a version of the isotropy statement that does characterize Leopoldt — one would want a condition on *all*  $T$  simultaneously, and since by Proposition 1.1 every projection rank is budgeted by  $r - \delta_p(K)$ , some aggregate of the image codimensions does see  $\delta_p$  and might work. And, given Remark 2.4, is the correspondence between the order of vanishing of  $E_p$  and the exceptional-zero count functorial enough to extend to nontrivial characters  $\chi$ , where the order becomes  $\#\{\mathfrak{p} \mid p : \chi(\mathfrak{p}) = 1\}$  rather than  $g$ ?

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